Test 1: Fall 2012 EIGHT PAGES NAME_____

The sheet resistance is $10^4$ ohms per square for a turned-on NMOS and use double that value for PMOS. Cg=10-14 Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use VT = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Tau_nmos_2=0.1ns. Tau_pmos_2=0.2ns.

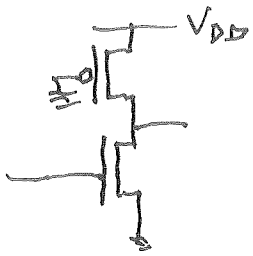
1. State two goals of the course. (5)

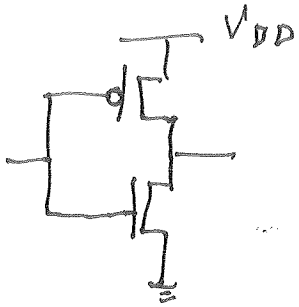
2. What is a die? (3)

3. Explain the lambda concept (3)

4. What transistor is normally in the resistive region when an inverter is pulling down? (4)

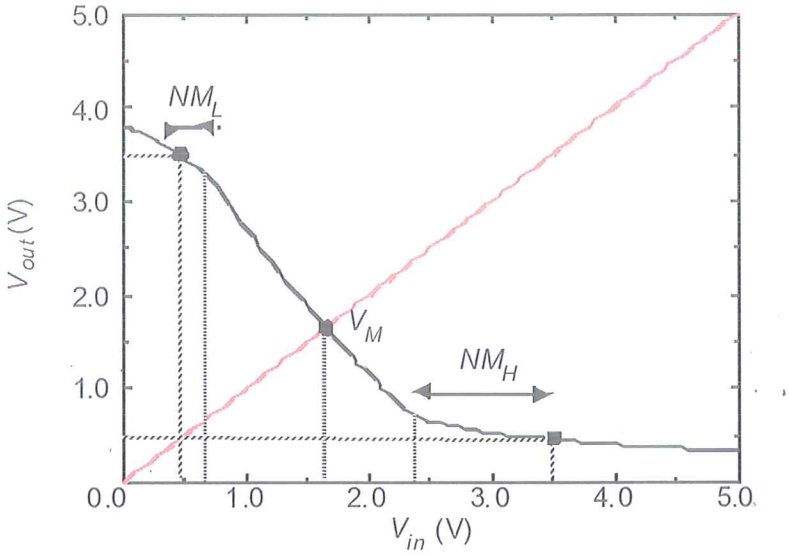
5. Label the two inverters as being switched or ratioed designs (6)



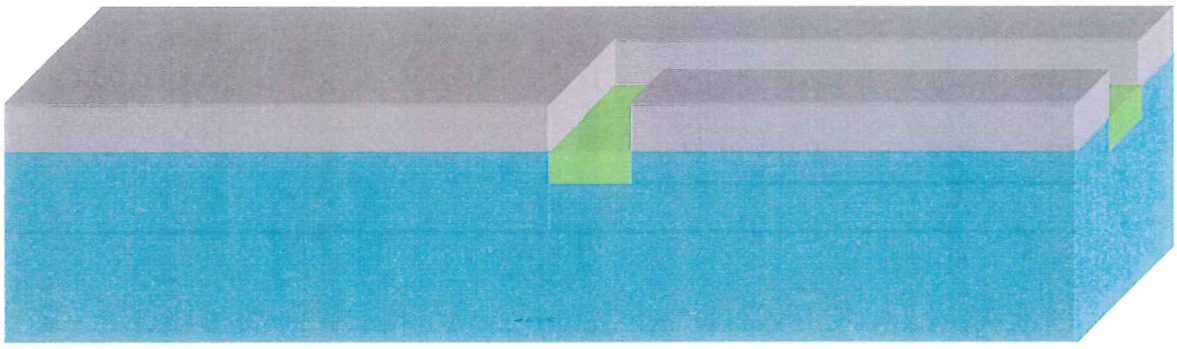




6. Why is noise margin important? (5)

7. What is the point labeled Vm? (4)




8. What is the green material?(4)

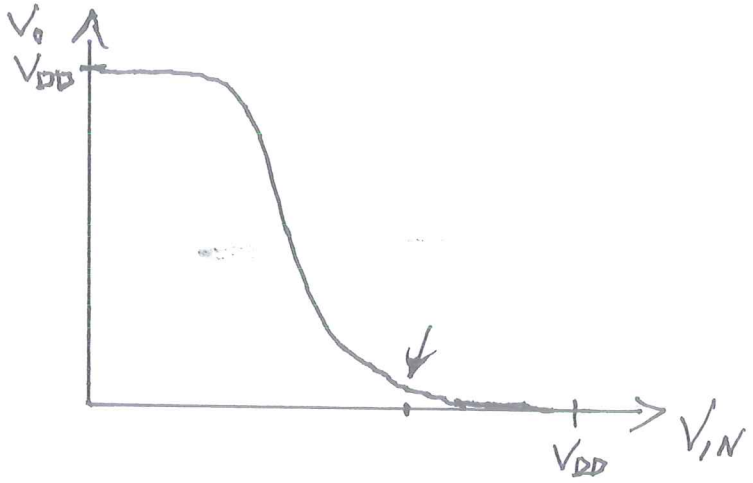


9. Why is an area called "active"? (5)

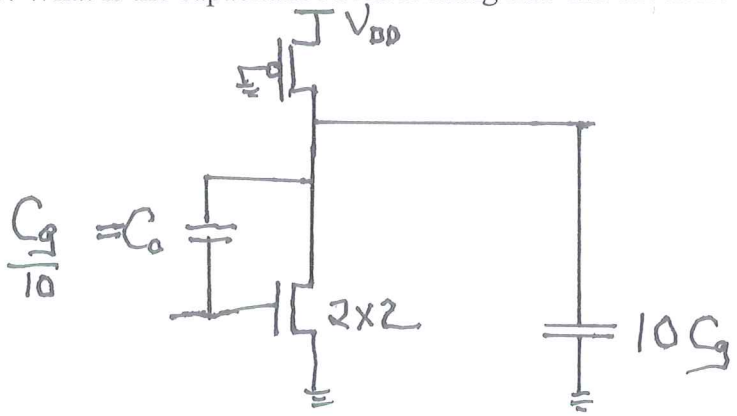
10. Why are some terms redundantly named, e.g. ndiff is sometimes called "green"(5)

11. Distinguish between the terms: well, nwell, pwell, bulk substrate doping, twin well process.(5) Write a short paragraph that describes each term as well as links them together in a coherent manner. You could do this in about six sentences.

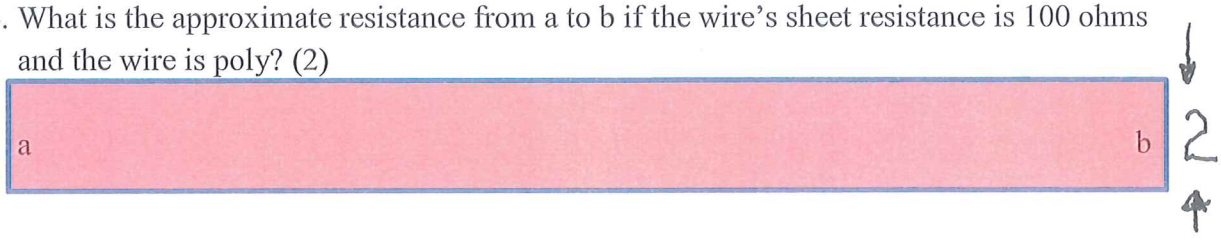
12. In what operating region would an nmos transistor be if Vin is at the voltage shown by the arrow (2). (This is a fully-complementary CMOS inverter.)




13. What is the capacitance seen looking into this inverter? (5)

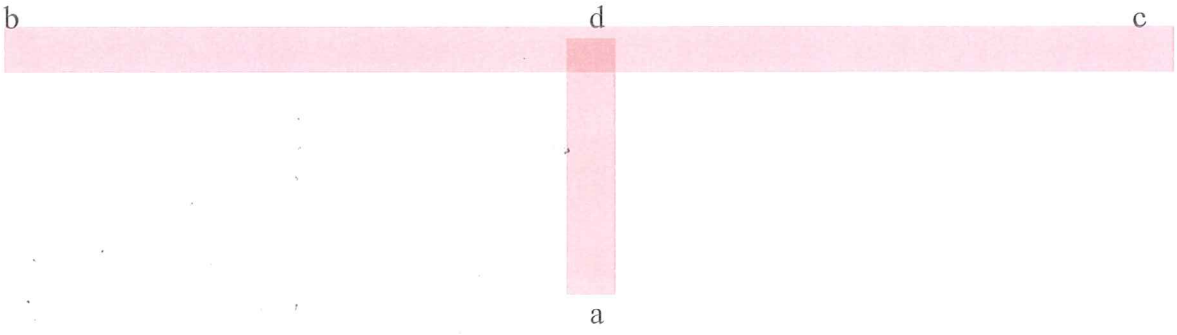



14. What is the approximate resistance from a to b if the wire's sheet resistance is 100 ohms and the wire is poly? (2)

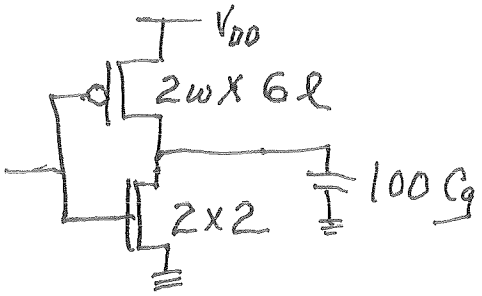


15. What is the capacitance of the wire in the previous problem? Assume a poly capacitance derating of Cg by a factor of 10.(3)

16. What is the Elmore delay from a to c? Show the extracted circuit, using symbols/labels, not numbers. Write the expression for the Elmore delay. (8 points)



17. What is the approximate rise time of this inverter? Use the "Tau model". (9)




same

18. What is approximately the rise time of this inverter using the RC approximation? (10)

19. A 1x inverter drives N-1 more inverters to form an optimal buffer chain to drive the final load of $3e^4$. What is N? (5)

20. An ideal fully-complementary clock source with a rise and fall time of 0.1 ns drives a 10,000 Cg load at a frequency of 10 MHz. What is the switching power *supplied by* the source to the load? (5)

21. What is velocity saturation in short channel devices? (2)

Test 1: Fall 2012 EIGHT PAGES NAME Allyn

The sheet resistance is $10^4$ ohms per square for a turned-on NMOS and use double that value for PMOS. Cg=10-14 Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use VT = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Tau_nmos_2=0.1ns. Tau_pmos_2=0.2ns.

1. State two goals of the course. (5)

1. Learn how to use a VLSI design tool.
2. Learn VLSI design techniques
⋮
more

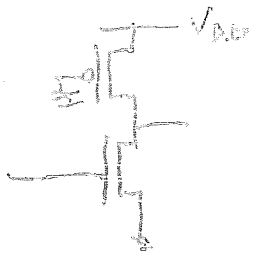
2. What is a die? (3) It is the small ~ 2mm → 1.5cm piece of silicon that is "the circuit."

3. Explain the lambda concept (3) The idea is that one could design using λ units & later, assign a value to put the layout into SI units. The size of λ is supposed to go down with time.

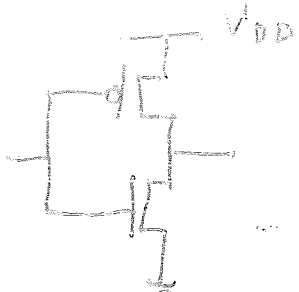
4. What transistor is normally in the resistive region when an inverter is pulling down? (4)

NMOS

5. Label the two inverters as being switched or ratioed designs (6)



ratioed

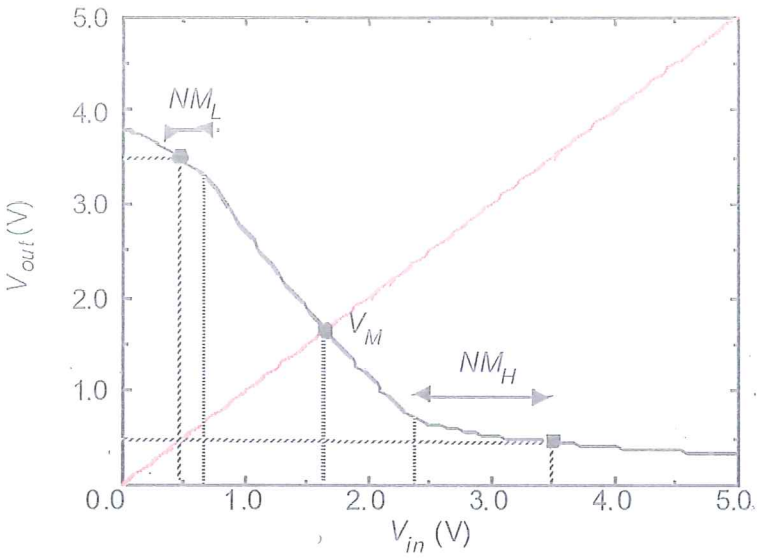


switched

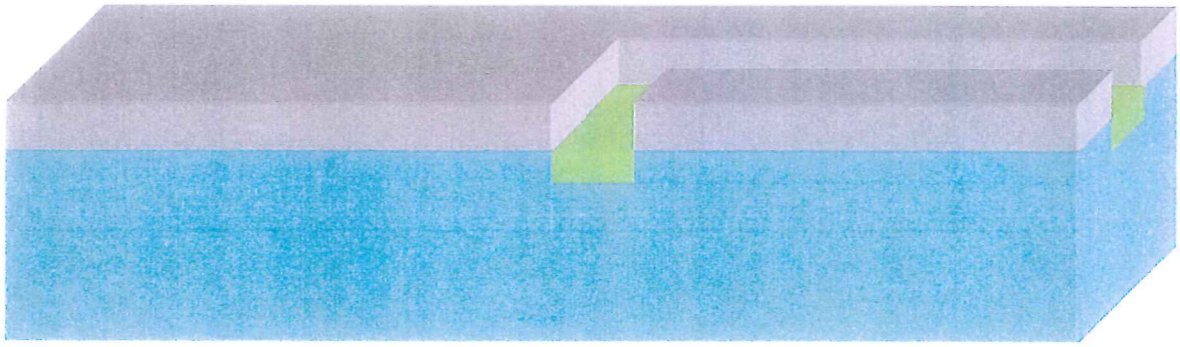
6. Why is noise margin important? (5) Noise margin is your voltage "comfort zone" where you can safely maintain a value that is a zero or a one.

7. What is the point labeled Vm? (4)

switching point
&/or
sat-sat




8. What is the green material?(4)



n-diff

9. Why is an area called "active"? (5) It is a region where the silicon surface is exposed so it can be n_doped, p_doped, or channel. See Prob. #8

10. Why are some terms redundantly named, e.g. ndiff is sometimes called "green"(5)

Different design tools developed over the years used different terminology. Names were equated to allow for upward (or sideways) compatibility.

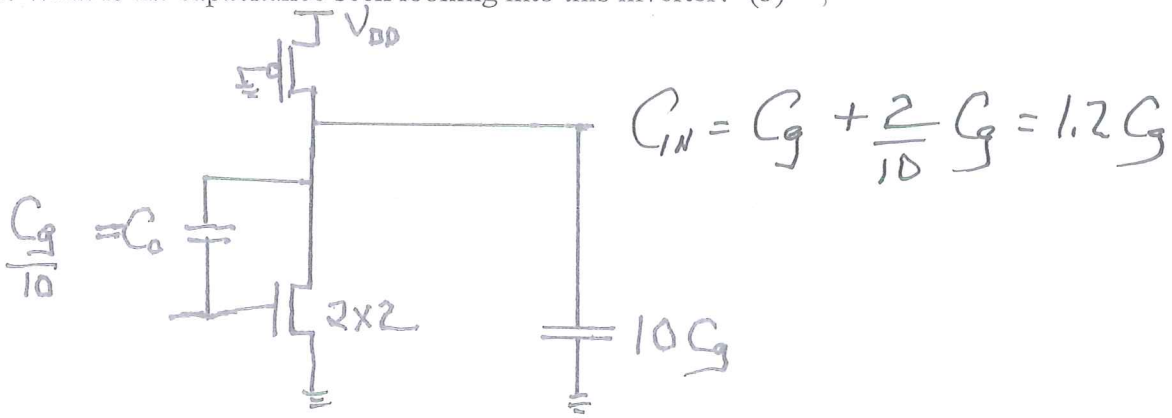
11. Distinguish between the terms: well, nwell, pwell, bulk substrate doping, twin well process.(5) Write a short paragraph that describes each term as well as links them together in a coherent manner. You could do this in about six sentences.

N&P_MOS transistors have a 4th terminal which is a connection to the substrate beneath them. NMOS needs a p_substrate. PMOS is reversed. The entire wafer could be all p_doped or all n_doped. In this case wells (or islands) can be inserted so N & PMOS have the proper substrate. In a twin well process, the wafer is not doped & n_wells and p_wells are both inserted.

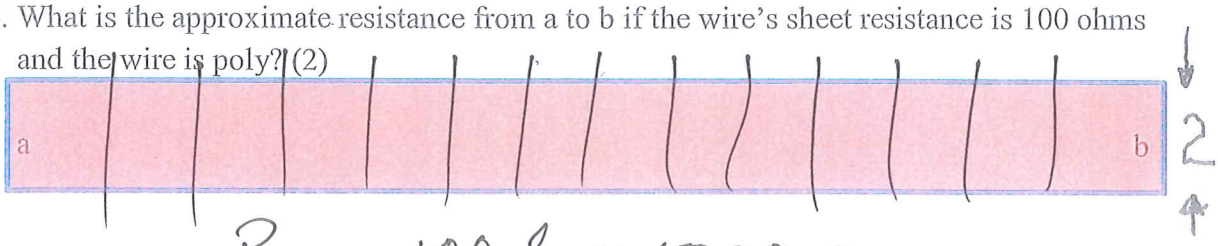
12. In what operating region would an nmos transistor be if Vin is at the voltage shown by the arrow (2). (This is a fully-complementary CMOS inverter.)




13. What is the capacitance seen looking into this inverter? (5)




14. What is the approximate resistance from a to b if the wire's sheet resistance is 100 ohms and the wire is poly? (2)



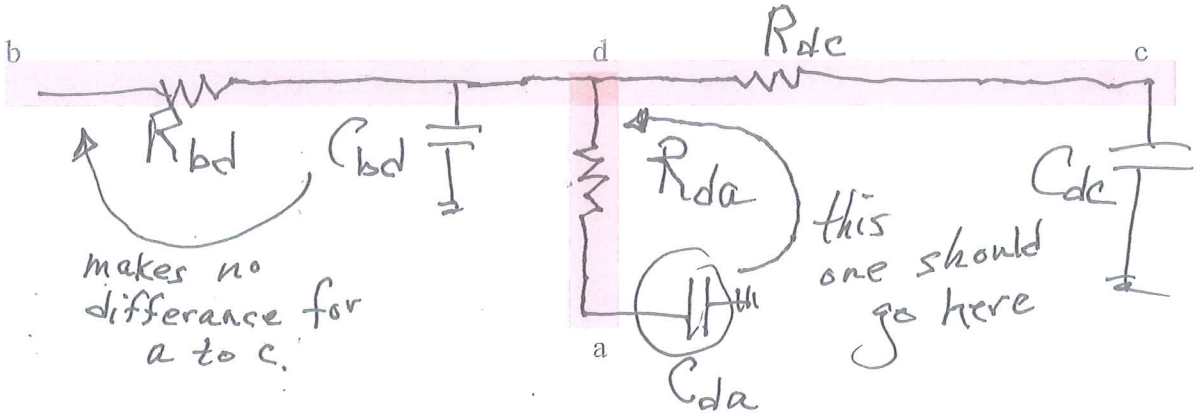

$R_{ab} = 100 \frac{l}{w} \approx 1300\ \Omega$

15. What is the capacitance of the wire in the previous problem? Assume a poly capacitance derating of Cg by a factor of 10.(3)

$$C = 3.13 \cdot \frac{1}{4} \cdot \frac{1}{10} = \frac{39}{40} C_g \sim C_g$$

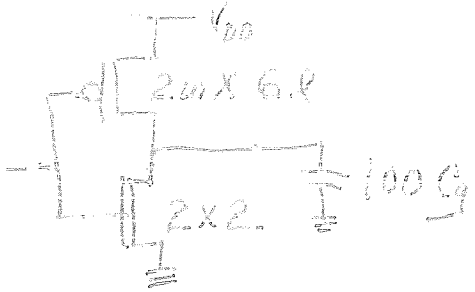
(Area)

16. What is the Elmore delay from a to c? Show the extracted circuit, using symbols/labels, not numbers. Write the expression for the Elmore delay. (8 points)




$$\text{Elmore a to c} = R_{da}(C_{da} + C_{bd} + C_{dc}) + R_{dc} C_{dc}$$

17. What is the approximate rise time of this inverter? Use the "Tau model". (9)




length

$$T_{01} = 0.2ns\left(\frac{C_L}{C_{pmos}} + \delta\right)\left(\frac{6^2}{2^2}\right)$$

$$= 1.8ns\ \frac{100C_g}{3C_g} + \delta \quad \text{forget } \delta < 1$$

$$\approx .6 \times 100\ ns$$

$$= 60\ ns$$

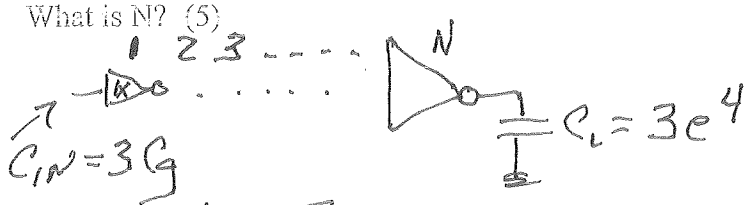
same

18. What is approximately the rise time of this inverter using the RC approximation? (10)

$$T_{01} = \underbrace{20 \times 10^3}_{R}\ \frac{6}{2} \cdot \underbrace{100C_g}_{C} = 10^4 \cdot 6 \cdot 10^{-12}\ s$$

$$= 6 \times 10^{-8} = 60ns$$

19. A 1x inverter drives N-1 more inverters to form an optimal buffer chain to drive the final load of $3e^4$. What is N? (5)



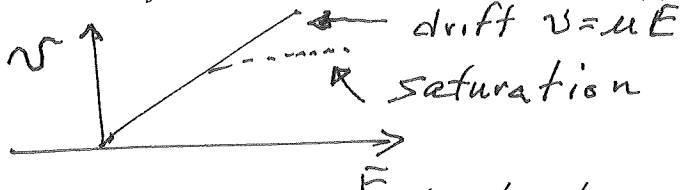

$$e = \sqrt[N]{\frac{3e^4}{3C_g}} \Rightarrow N=4$$

fast: OK, $CV^2f$ works

20. An ideal fully-complementary clock source with a rise and fall time of 0.1 ns drives a 10,000 Cg load at a frequency of 10 MHz. What is the switching power *supplied by* the source to the load? (5)

$$P_{switch} = CV^2f = 10^4 \times 10^{-14} \cdot 2.5 \cdot 10^7$$
$$= 25 \times 10^{-14+11} = 25 \times 10^{-3} W$$

21. What is velocity saturation in short channel devices? (2)




If the channel gets to be very short & $V_{DD}$ is still ~1V (due to noise & threshold and other considerations) the E field [V/m] becomes very high. The carriers' speed can no longer increase

Test 1: Fall 2012 EIGHT PAGES NAME Grading Key

The sheet resistance is $10^4$ ohms per square for a turned-on NMOS and use double that value for PMOS. Cg=10-14 Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use VT = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Tau_nmos_2=0.1ns. Tau_pmos_2=0.2ns.

1. State two goals of the course. (5)

look for broad goals

2. What is a die? (3)

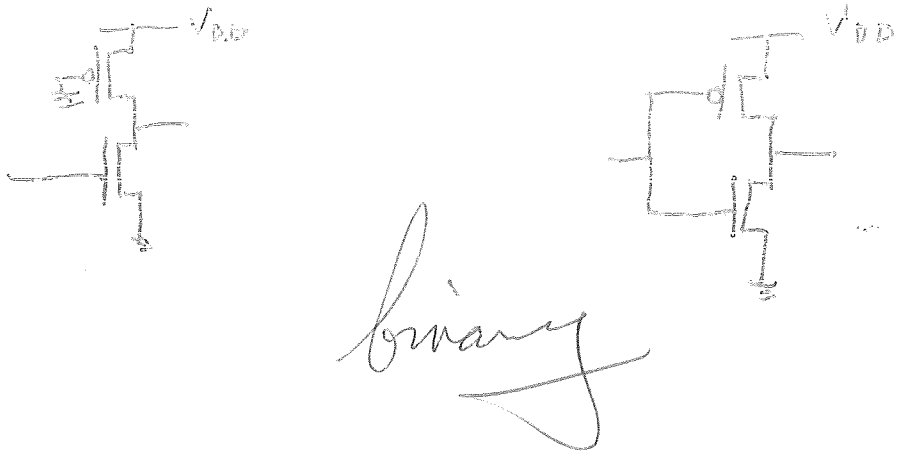
3. Explain the lambda concept (3)

scaling !!!

4. What transistor is normally in the resistive region when an inverter is pulling down? (4)

biasing

5. Label the two inverters as being switched or ratioed designs (6)

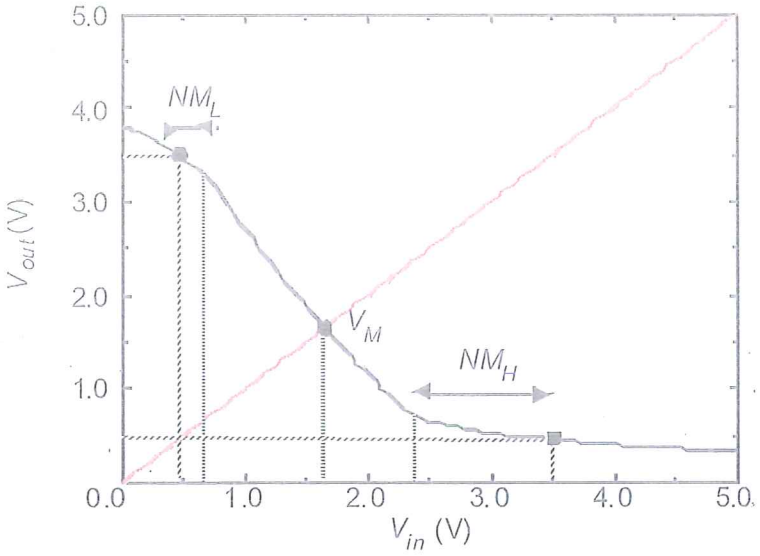


biasing

6. Why is noise margin important? (5)

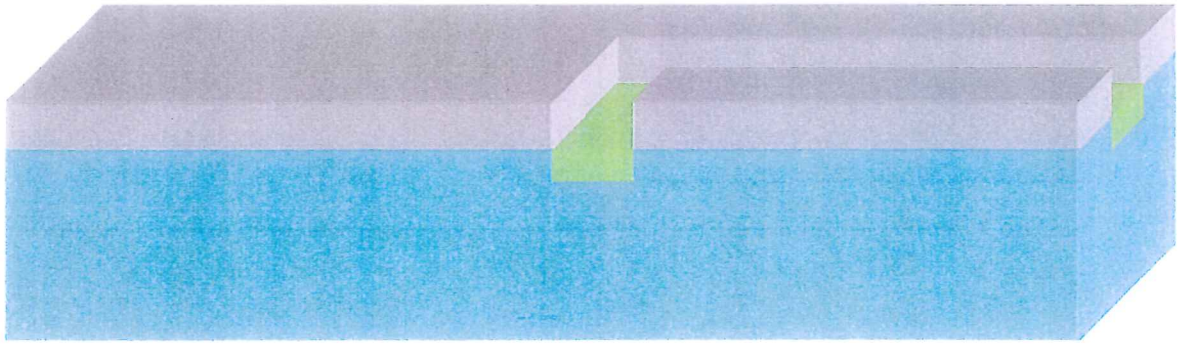
variable

7. What is the point labeled Vm? (4)




also right answers

8. What is the green material?(4)



mostly binary

9. Why is an area called "active"? (5)

variable. No one "nailed" it.

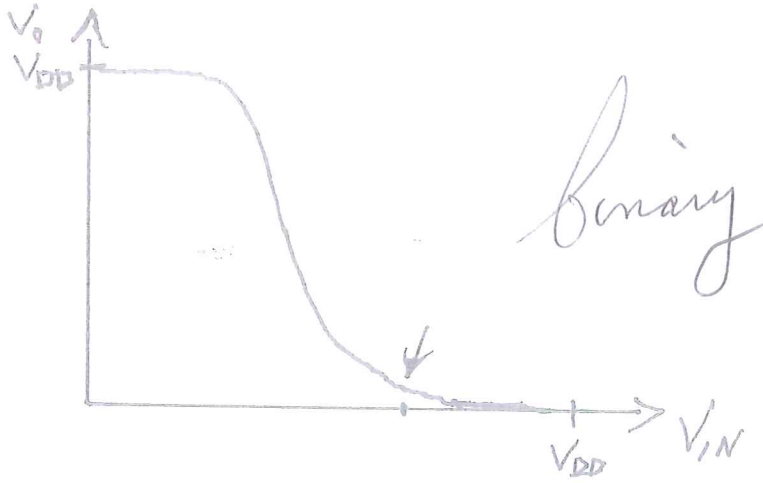
10. Why are some terms redundantly named, e.g. ndiff is sometimes called "green"(5)

Students missed the question & gave trivial answer, basically stating the facts of the question. Some got closer to the "concept." Usually -2 in that case

11. Distinguish between the terms: well, nwell, pwell, bulk substrate doping, twin well process.(5) Write a short paragraph that describes each term as well as links them together in a coherent manner. You could do this in about six sentences.

Took for 5 right answers @ 1pt. each.

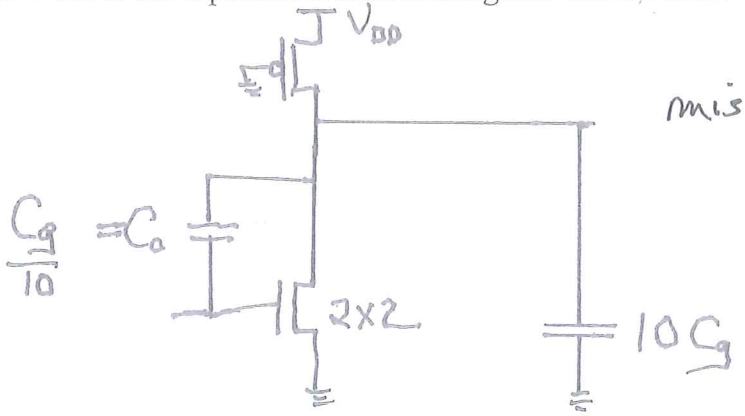
12. In what operating region would an nmos transistor be if Vin is at the voltage shown by the arrow (2). (This is a fully-complementary CMOS inverter.)




binary

13. What is the capacitance seen looking into this inverter? (5)

if add load (-3)

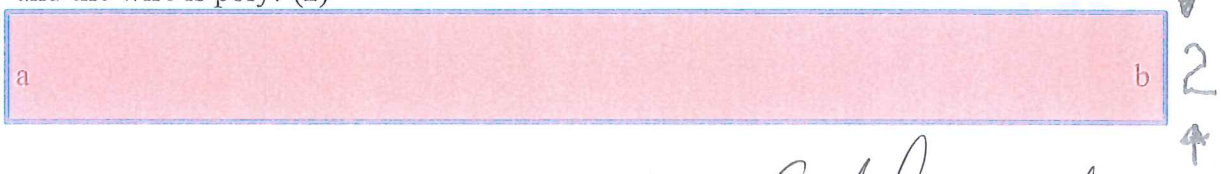



missed $C_m$ (-2)

see pMOS (-2)

only $C_m$ @ $\frac{C_g}{10}$ and no nmos gate (-5)

14. What is the approximate resistance from a to b if the wire's sheet resistance is 100 ohms and the wire is poly? (2)




~ 500 - 1300 OK for off (-1)

(ask for ohms here. If started taking off -2.)

15. What is the capacitance of the wire in the previous problem? Assume a poly capacitance derating of Cg by a factor of 10.(3)

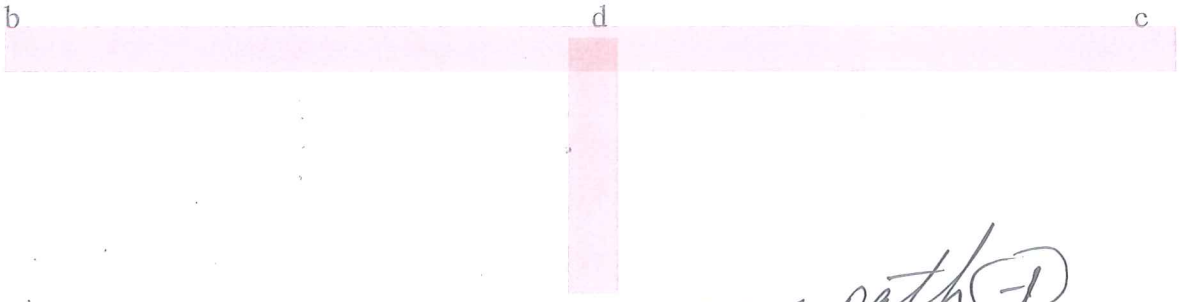
X2 (-1)

Check back to #14 for consistency.

no derating (-1) or wrong way i.e. X10

÷4 (-1)

16. What is the Elmore delay from a to c? Show the extracted circuit, using symbols/labels, not numbers. Write the expression for the Elmore delay. (8 points)



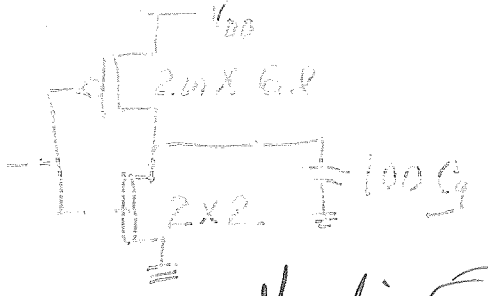

wrong path (-1)

extraction worth 4. (-1) each wire

Expression (-1) each missing term.

Cg's at the end (-1)

17. What is the approximate rise time of this inverter? Use the "Tau model". (9)




arithmetic (-1)

3 each

$0.2\left(\frac{6}{2}\right)^2 \frac{100C_g}{3C_g}$ ns

$0.2 \cdot \frac{(3.2 \cdot 6) \cdot 10,0}{4\ \ 3}$

$= 0.2 \times \frac{12}{4} \times 100$ ns

$= \frac{2.4}{4} \times 100 = 60$ ns

same

18. What is approximately the rise time of this inverter using the RC approximation? (10)

5 $R = 2\times10^4 \cdot \frac{6}{2} = 6\times10^4$ formula wrong (-2)

5 $C_L = \frac{10^{-14} \cdot 12 - 3\times10^{-14}}{4}$ $10^{-12}$

used $R_{pnmos}$ (-2)

$RC_L = 6\times10^{-8} = 60$ ns

(-1) arithmetic

R uses $\frac{W}{L}$ (-2)

add $C_{gmos}$ to Load (-2)

No $C_L$ at all (-5)

19. A 1x inverter drives N-1 more inverters to form an optimal buffer chain to drive the final load of $3e^4$. What is N? (5)

another x3 ⇒ N=5 (-1)

x3 problems (-1)

if no Cg, & got answer, forgot it.

counting → N=5, (-0), all wrong -5

N=e! = 4 No work (-5) A.W. or is it obvious??

20. An ideal fully-complementary clock source with a rise and fall time of 0.1 ns drives a 10,000 Cg load at a frequency of 10 MHz. What is the switching power *supplied by* the source to the load? (5)

Arithmetic (-1)

Zero (-5)

load = 3Cg (-3)

~~X3~~

calculated f from R(&F) (-2)

NO Answer -5, some work -4,

21. What is velocity saturation in short channel devices? (2)